IT114 Project Milestone 1

Student submission

Assignment revision: 2 | Maximum points: 10 | Student: aia47 | Submitted: 2026-11-10T06:58:36.481Z | Submission revision: 117 | Receipt: v4r_01M7240RT1Z1KHS1D96NVAXA78 | Exported: 2026-11-10T06:59:13.186Z

Worksheet link

https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-project-milestone-1/aia47

Grading link

https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-project-milestone-1/grading/aia47

8 submission warning(s) were acknowledged.

AI disclosure: Completed; 1 entry; 0 declared AI use; 1 declared no AI use; policy v1

Server startup

Demonstrate that the server starts from the repository root, listens, and accepts connections.

Server source

URL prompt

Provide direct GitHub file URLs for the server entry point and connection-accepting code on `Milestone1`; add multiple URLs if those are in separate files.

Requirements and potential deductions

- The linked source starts a listener and accepts incoming connections. (up to -0.55 points)
- Relevant code comments identify the student's UCID, date, and purpose. (up to -0.2 points)

Warning at submission: url-path

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Server/Server.java

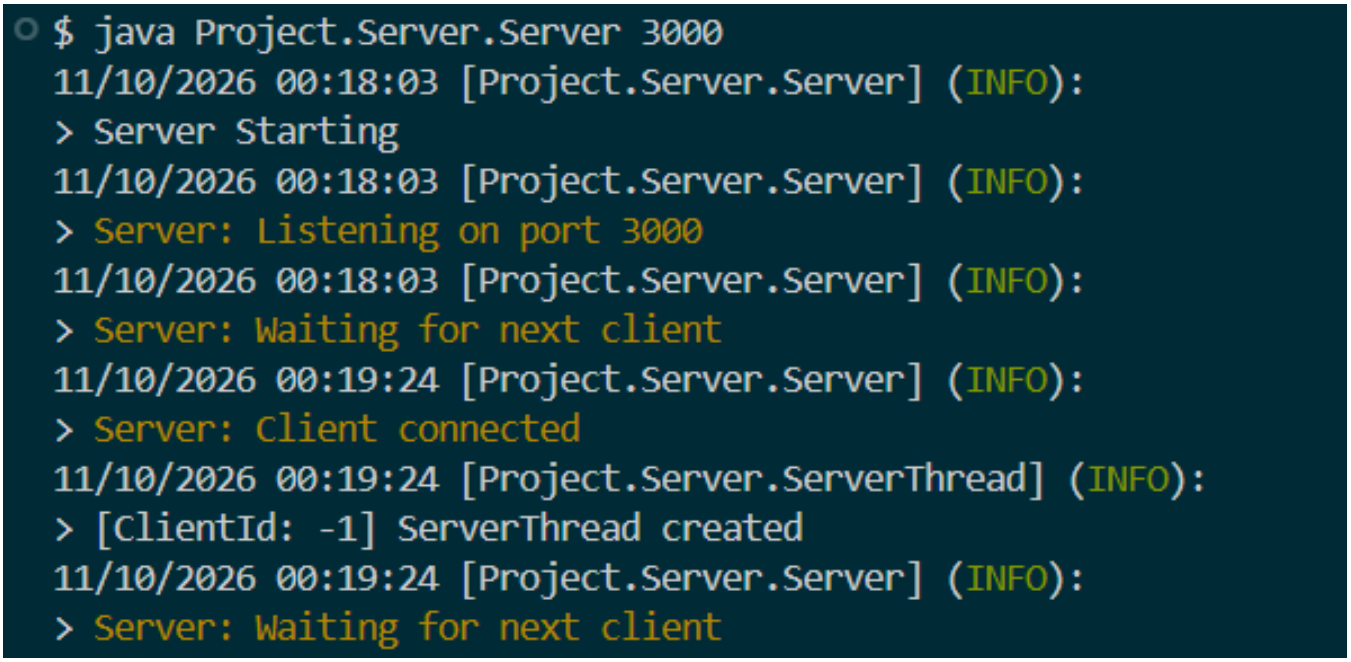
Listening output

Image evidence prompt

Upload a readable terminal capture showing the server started and listening from the repository root.

Requirements and potential deductions

- The server process visibly starts and remains ready for connections. (up to -0.5 points)



Accept explanation

Response prompt

Explain which method waits for a connection and how your server passes an accepted client to its connection handler.

Requirements and potential deductions

- The response connects the listener, accept call, and handler creation in the submitted source. (up to -0.25 points)

Submitted response

Start() waits for a client with serverSocket.accept() making a new serverThread that hadles the client

Multiple clients

Keep at least three clients connected to one server at the same time.

Connection source

URL prompt

Provide direct GitHub file URLs for the server's client registry or callback and the per-client handler on `Milestone1`.

Requirements and potential deductions

- The server keeps independent connections and tracks connected clients. (up to -0.75 points)

Warning at submission: url-path

Warning at submission: url-path

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Server/ServerThread.java
https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Server/Server.java

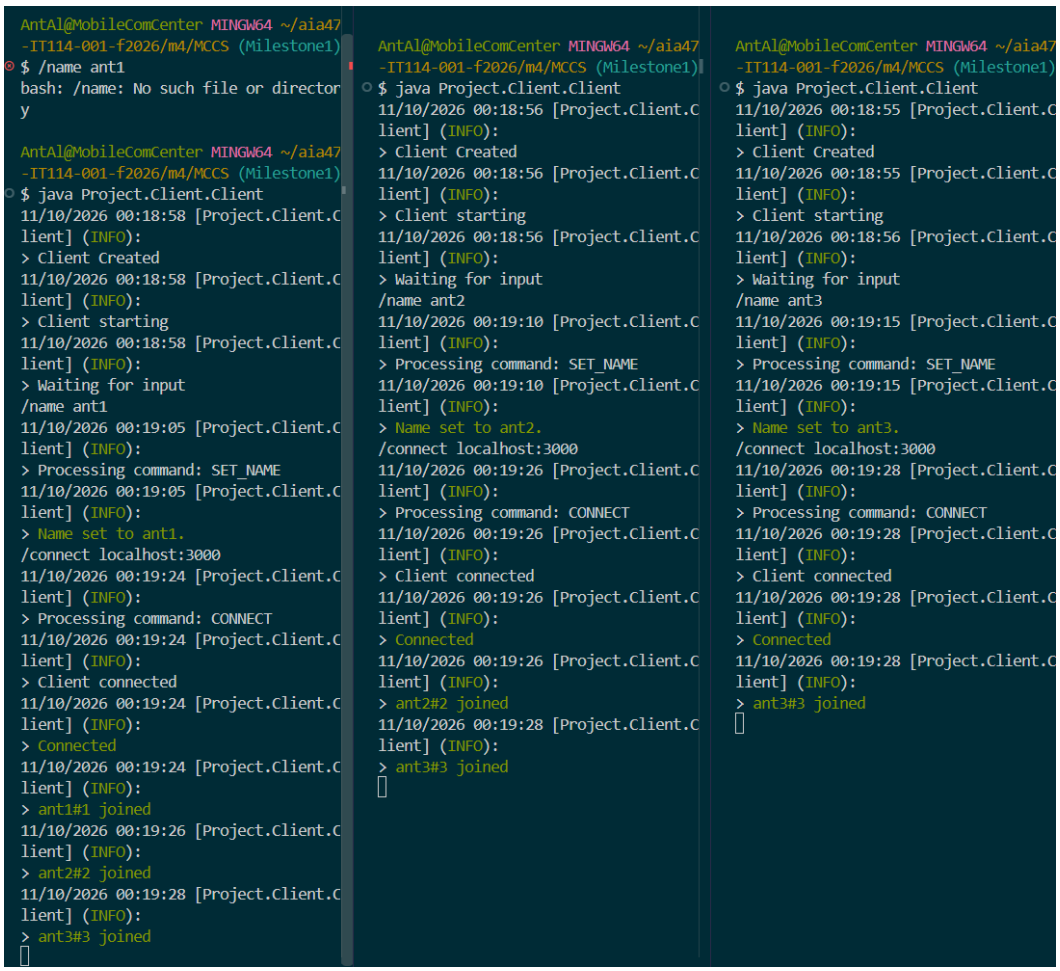
Three-client output

Image evidence prompt

Upload readable terminal captures showing one server and at least three named clients connected simultaneously; split terminals or multiple images are fine.

Requirements and potential deductions

- Three distinct clients are concurrently connected to the same running server. (up to -0.75 points)



Connection explanation

Response prompt

Explain how the submitted server code handles multiple clients without one client's input blocking the others.

Requirements and potential deductions

- The response identifies the actual per-client execution and server tracking in the linked source. (up to -0.5 points)

Submitted response

They each are handled by a different serverthread and keeps track of connected

Client identity and connection

Each of three clients sets a name, connects from the command line, and receives connection confirmation.

Client source

URL prompt

Provide direct GitHub file URLs for `/name`, `/connect`, and connection confirmation in the `Project/Client/` code on `Milestone1`.

Requirements and potential deductions

- The client accepts a name before opening a connection and sends that identity to the server. (up to -0.5 points)
- The client processes the server's assigned identity and reports connection confirmation. (up to -0.25 points)

Warning at submission: url-path

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Client/Client.java

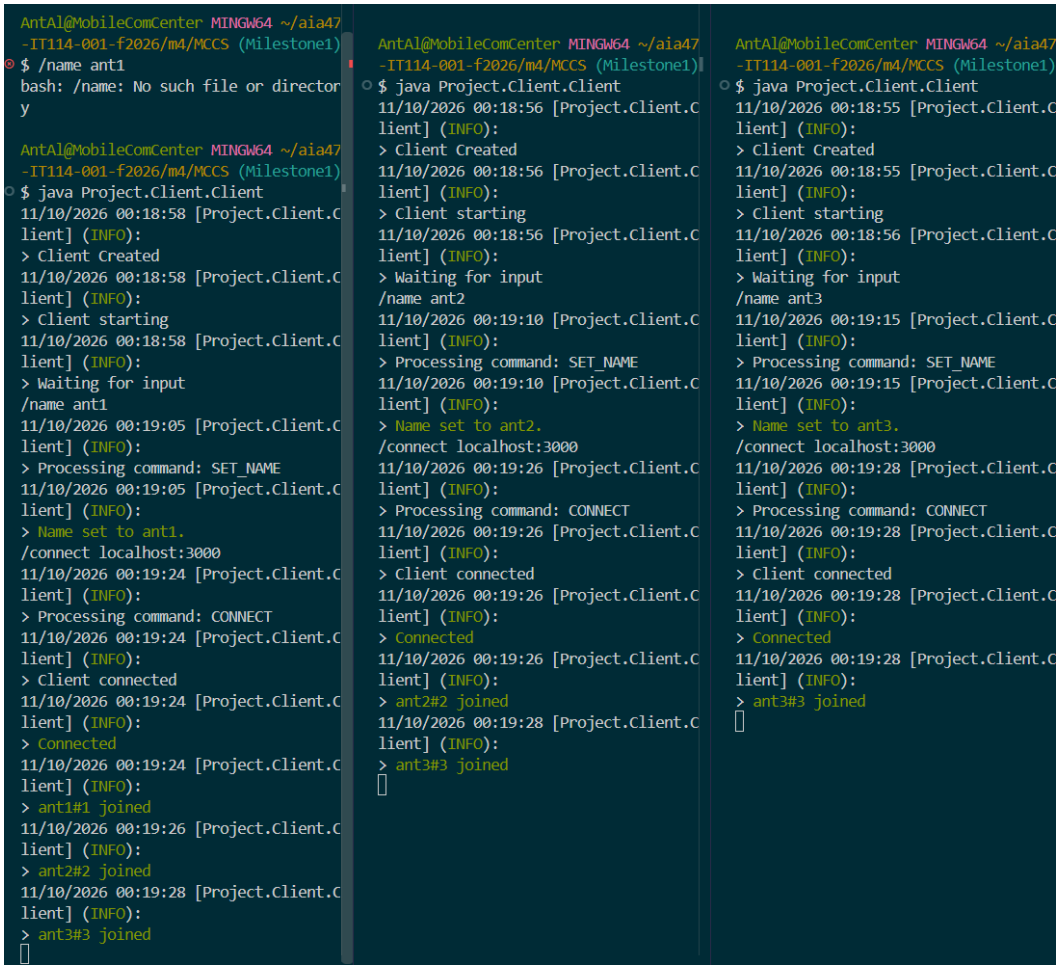
Three-client command output

Image evidence prompt

Upload readable captures of `/name` and `/connect` for each of three clients, including each client's successful connection confirmation.

Requirements and potential deductions

- All three named clients complete `/connect` and show their own confirmation. (up to -0.5 points)



Identity explanation

Response prompt

Explain how your `/name` and `/connect` input reaches the server and how the assigned ID returns to the client.

Requirements and potential deductions

- The response connects command processing, identity sending, server assignment, and client confirmation. (up to -0.25 points)

Submitted response

/name saves name in client and /connect sends name to server. Server sends back assigned id to client who saves it.

Message exchange

Each connected client can send ordinary chat text that reaches the other clients.

Message source

URL prompt

Provide direct GitHub file URLs for client sending and the server's message routing on `Milestone1`.

Requirements and potential deductions

- The client sends typed message data and the server distributes it to connected clients. (up to -0.75 points)

Warning at submission: url-path

Warning at submission: url-path

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Server/Server.java
https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Client/Client.java

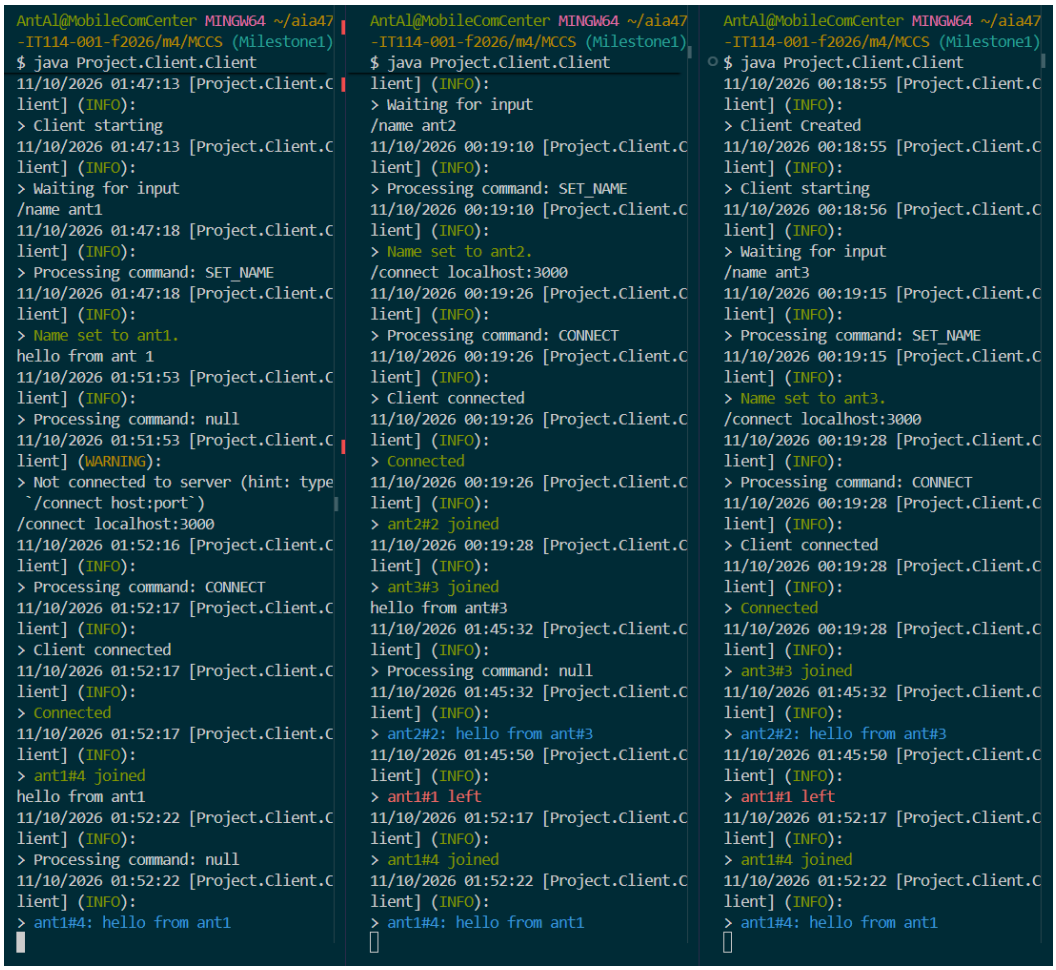
Message output

Image evidence prompt

Upload readable captures of a different ordinary message sent by each of the three clients, with the received messages visible on the other clients.

Requirements and potential deductions

- Each of the three clients sends a message received by the other connected clients. (up to -0.75 points)



Message explanation

Response prompt

Trace one submitted message from client send through server routing to a receiving client.

Requirements and potential deductions

- The response names the actual send, process, and receive steps in the linked code. (up to -0.5 points)

Submitted response

client sends a message with sendMessage to the server which uses handleMessage. The server broadcost sends it to every c
and they print it in processMessage.

Disconnect and reconnect

A client can leave without stopping the server; after server shutdown and restart, clients can reconnect.

Cleanup source

URL prompt

Provide direct GitHub file URLs for client disconnect handling and server connection cleanup on `Milestone1`.

Requirements and potential deductions

- Client and server cleanup close the affected connection and remove its tracked identity without ending unrelated clients. (up to -0.75 points)

Warning at submission: url-path
Warning at submission: url-path

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Server/Server.java
https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/Milestone1/M4/MCCS/Project/Client/Client.java

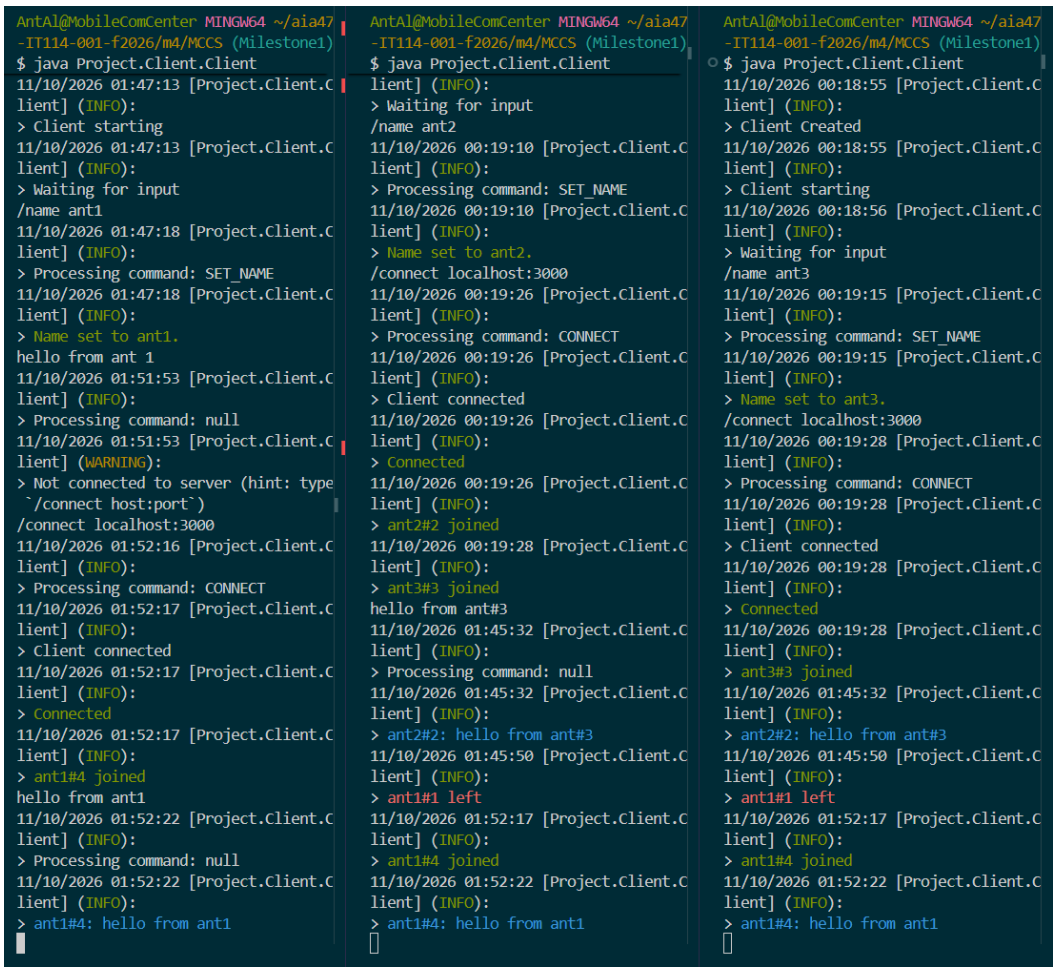
Reconnection output

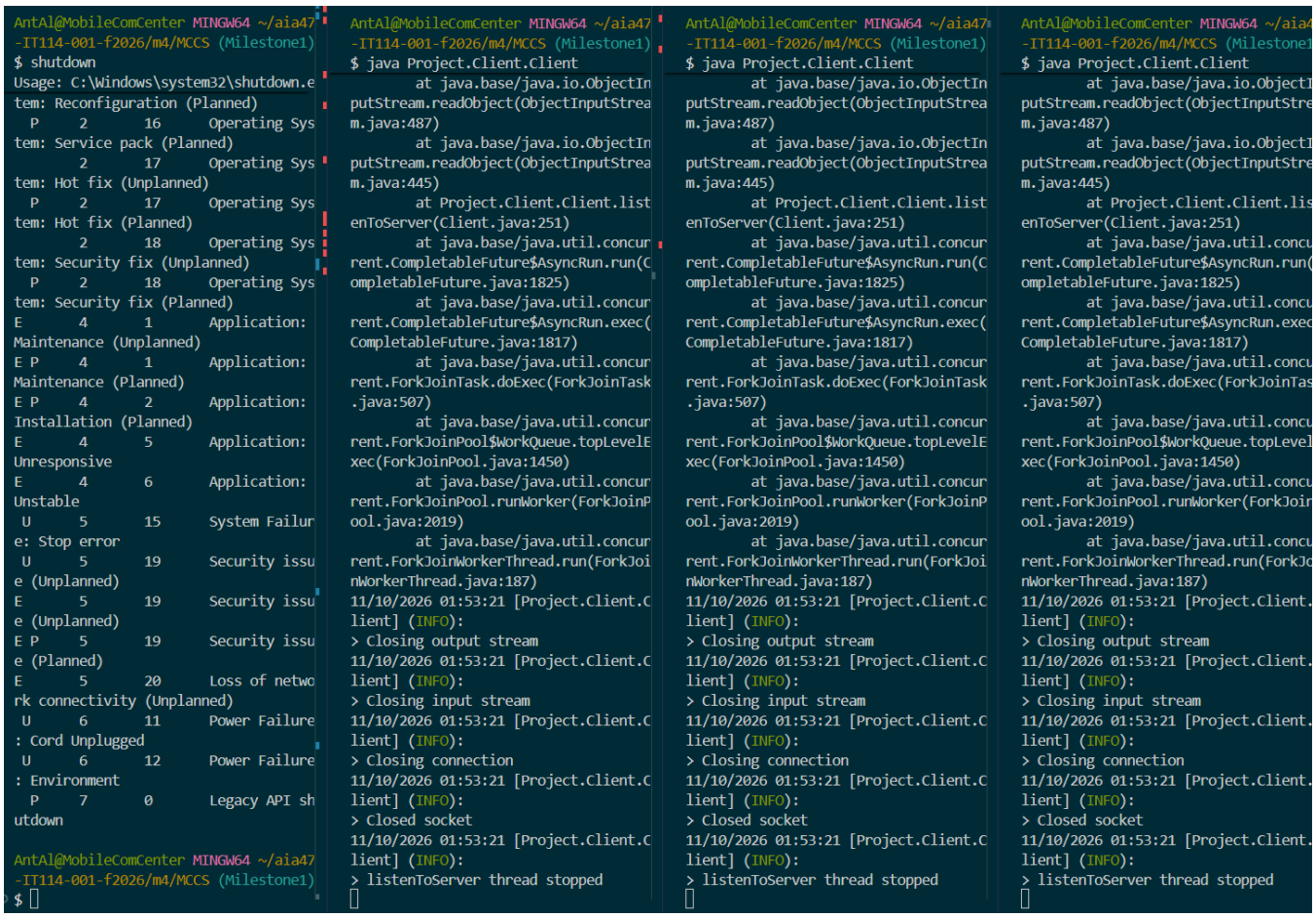
Image evidence prompt

Upload readable terminal captures showing one client disconnect while the server and another client remain active, server shutdown while clients remain open but disconnected, and a successful client reconnect after server restart.

Requirements and potential deductions

- One client leaves while another keeps using the running server. (up to -0.25 points)
- Server shutdown leaves clients running but disconnected. (up to -0.25 points)
- A client reconnects after server restart. (up to -0.25 points)


AntAl@MobileComCenter MINGW64 ~/aia47-IT114-001-f2026/m4/MCCS (Milestone1)
$ java Project.Client.Client
11/10/2026 01:47:13 [Project.Client.Client] (INFO):
> Client starting
11/10/2026 01:47:13 [Project.Client.Client] (INFO):
> Waiting for input
/name ant1
11/10/2026 01:47:18 [Project.Client.Client] (INFO):
> Processing command: SET_NAME
11/10/2026 01:47:18 [Project.Client.Client] (INFO):
> Name set to ant1.
hello from ant 1
11/10/2026 01:51:53 [Project.Client.Client] (INFO):
> Processing command: null
11/10/2026 01:51:53 [Project.Client.Client] (WARNING):
> Not connected to server (hint: type `/connect host:port`)
/connect localhost:3000
11/10/2026 01:52:16 [Project.Client.Client] (INFO):
> Processing command: CONNECT
11/10/2026 01:52:17 [Project.Client.Client] (INFO):
> Client connected
11/10/2026 01:52:17 [Project.Client.Client] (INFO):
> Connected
11/10/2026 01:52:17 [Project.Client.Client] (INFO):
> ant1#4 joined
hello from ant1
11/10/2026 01:52:22 [Project.Client.Client] (INFO):
> Processing command: null
11/10/2026 01:52:22 [Project.Client.Client] (INFO):
> ant1#4: hello from ant1
AntAl@MobileComCenter MINGW64 ~/aia47-IT114-001-f2026/m4/MCCS (Milestone1)
$ java Project.Client.Client
11/10/2026 00:19:10 [Project.Client.Client] (INFO):
> Waiting for input
/name ant2
11/10/2026 00:19:10 [Project.Client.Client] (INFO):
> Processing command: SET_NAME
11/10/2026 00:19:10 [Project.Client.Client] (INFO):
> Name set to ant2.
/connect localhost:3000
11/10/2026 00:19:26 [Project.Client.Client] (INFO):
> Processing command: CONNECT
11/10/2026 00:19:26 [Project.Client.Client] (INFO):
> Client connected
11/10/2026 00:19:26 [Project.Client.Client] (INFO):
> Connected
11/10/2026 00:19:26 [Project.Client.Client] (INFO):
> ant2#2 joined
11/10/2026 00:19:28 [Project.Client.Client] (INFO):
> ant3#3 joined
hello from ant#3
11/10/2026 01:45:32 [Project.Client.Client] (INFO):
> Processing command: null
11/10/2026 01:45:32 [Project.Client.Client] (INFO):
> ant2#2: hello from ant#3
11/10/2026 01:45:50 [Project.Client.Client] (INFO):
> ant1#1 left
11/10/2026 01:52:17 [Project.Client.Client] (INFO):
> ant1#4 joined
11/10/2026 01:52:22 [Project.Client.Client] (INFO):
> ant1#4: hello from ant1
AntAl@MobileComCenter MINGW64 ~/aia47-IT114-001-f2026/m4/MCCS (Milestone1)
$ java Project.Client.Client
11/10/2026 00:18:55 [Project.Client.Client] (INFO):
> Client Created
11/10/2026 00:18:55 [Project.Client.Client] (INFO):
> Client starting
11/10/2026 00:18:56 [Project.Client.Client] (INFO):
> Waiting for input
/name ant3
11/10/2026 00:19:15 [Project.Client.Client] (INFO):
> Processing command: SET_NAME
11/10/2026 00:19:15 [Project.Client.Client] (INFO):
> Name set to ant3.
/connect localhost:3000
11/10/2026 00:19:28 [Project.Client.Client] (INFO):
> Processing command: CONNECT
11/10/2026 00:19:28 [Project.Client.Client] (INFO):
> Client connected
11/10/2026 00:19:28 [Project.Client.Client] (INFO):
> Connected
11/10/2026 00:19:28 [Project.Client.Client] (INFO):
> ant3#3 joined
11/10/2026 01:45:32 [Project.Client.Client] (INFO):
> ant2#2: hello from ant#3
11/10/2026 01:45:50 [Project.Client.Client] (INFO):
> ant1#1 left
11/10/2026 01:52:17 [Project.Client.Client] (INFO):
> ant1#4 joined
11/10/2026 01:52:22 [Project.Client.Client] (INFO):
> ant1#4: hello from ant1



Cleanup explanation

Response prompt

Explain which client and server cleanup steps permit disconnect and reconnect without a stale client identity.

Requirements and potential deductions

- The response connects actual socket closing, registry removal, and new connection setup. (up to -0.5

points)

Submitted response

On disconnect the server removes the client from connectClients, annouces it left and closes the socket and streams. Since ol
gone when the client reconnects it gets a new ID.

Reflection

Pull request URL

URL prompt

Submit the pull-request URL from `Milestone1` into `main`.

Requirements and potential deductions

- The pull request contains the Milestone 1 source and remains open for the exported PDF commit. (up to -0.2 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/pull/6

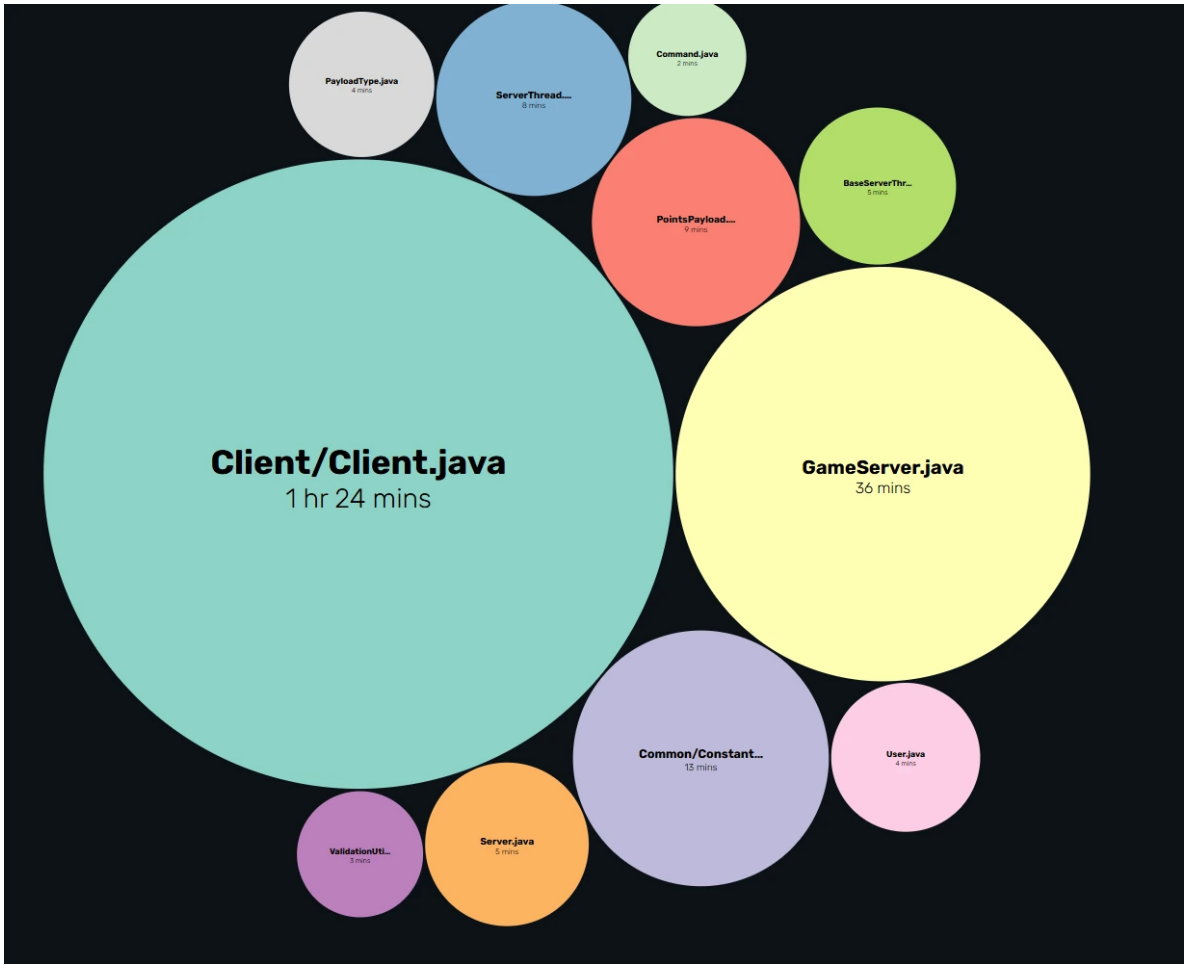
WakaTime project overview

Image evidence prompt

Upload a readable WakaTime project overview showing your repository name; time duration is not graded.

Requirements and potential deductions

- The capture identifies the project repository. (up to -0.1 points)



WakaTime file activity

Image evidence prompt

Upload a readable WakaTime file-activity capture for this milestone; time duration is not graded.

Requirements and potential deductions

- The capture identifies relevant Java file activity. (up to -0.1 points)



Files		Branches	
1 hr 24 mins	Project/Client/Client.java	3 hrs	Milestone1
36 mins	Project/Server/GameServer.java		
13 mins	Project/Common/Constants.java		
9 mins	Project/Common/PointsPayload.java		
8 mins	Project/Server/ServerThread.java		
5 mins	Project/Server/Server.java		
5 mins	Project/Server/BaseServerThread.java		
4 mins	Project/Common/User.java		
4 mins	Project/Common/PayloadType.java		
3 mins	Project/Common/ValidationUtils.java		
2 mins	Project/Client/Command.java		

Learned or verified

Response prompt

What did you learn or verify about the project framework and its connection behavior?

Requirements and potential deductions

- The response identifies a concrete lesson or verification result. (up to -0.2 points)

Submitted response

It helped better my understadning of serverthreads and the way ther servers handle clients.

Easiest part

Response prompt

Which part of this milestone was easiest, and why?

Requirements and potential deductions

- The response names a specific part and explains why it was easier. (up to -0.2 points)

Submitted response

The terminal commands were the easiest.

Difficult or uncertain part

Response prompt

Which part was most difficult or remains uncertain, and what did you try or check?

Requirements and potential deductions

- The response names a specific difficulty or uncertainty and a relevant check. (up to -0.2 points)

Submitted response

I had trouble understanding the instructions but hat was wholly due to my inability to properly read said instructions.